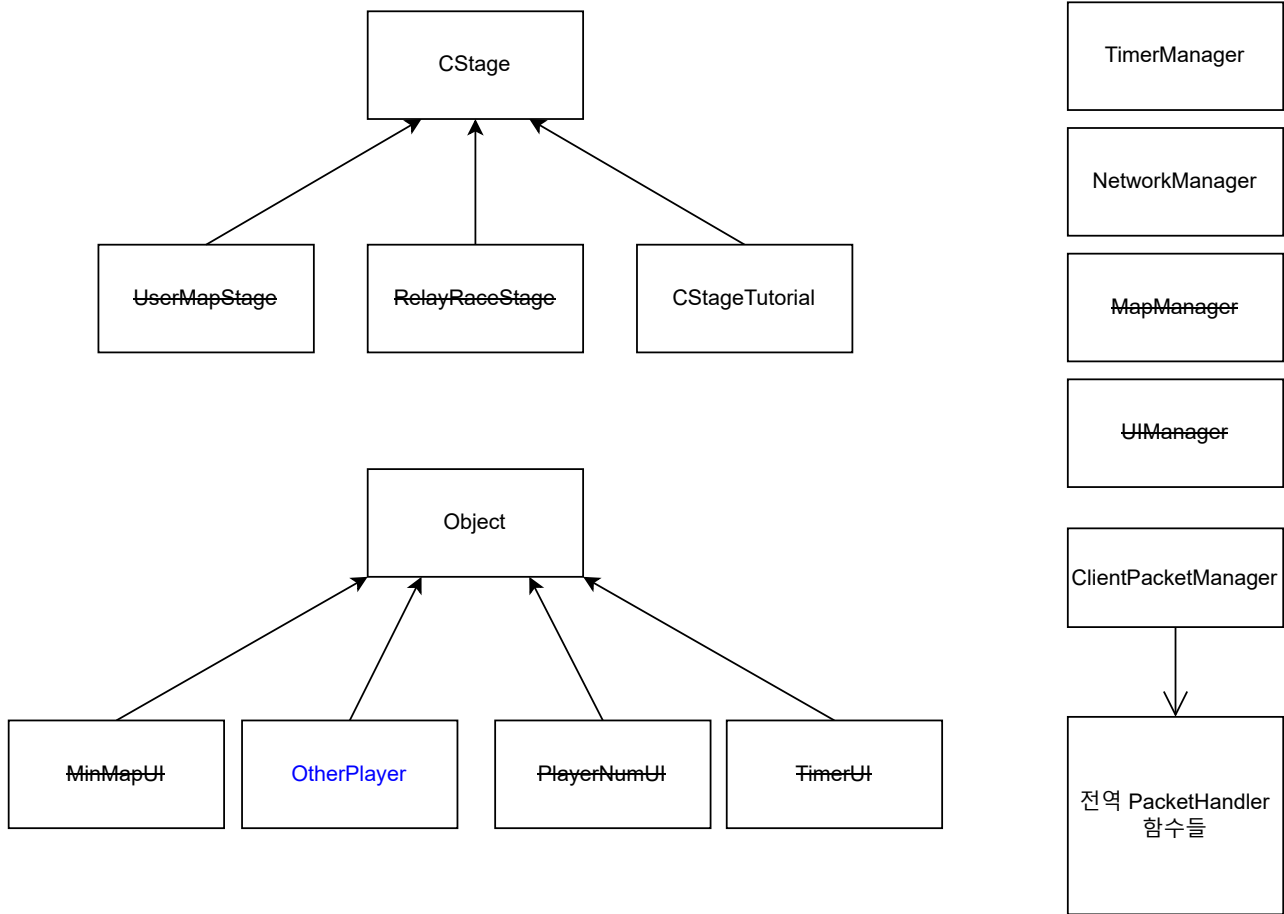
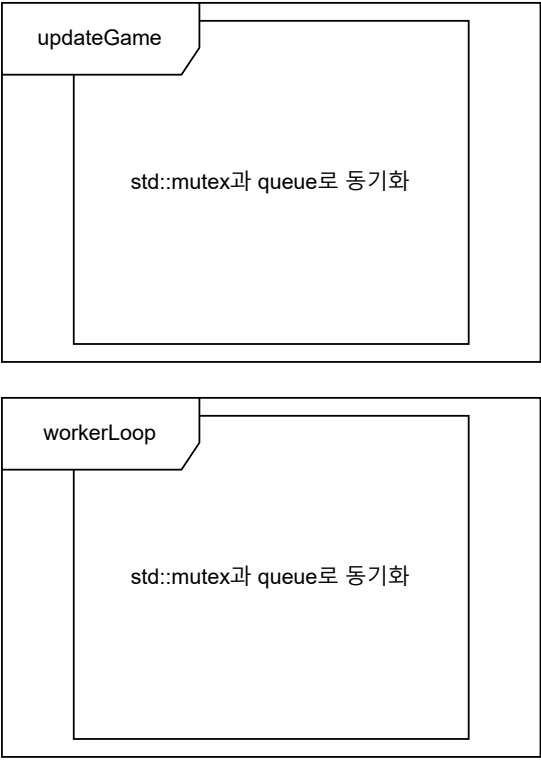
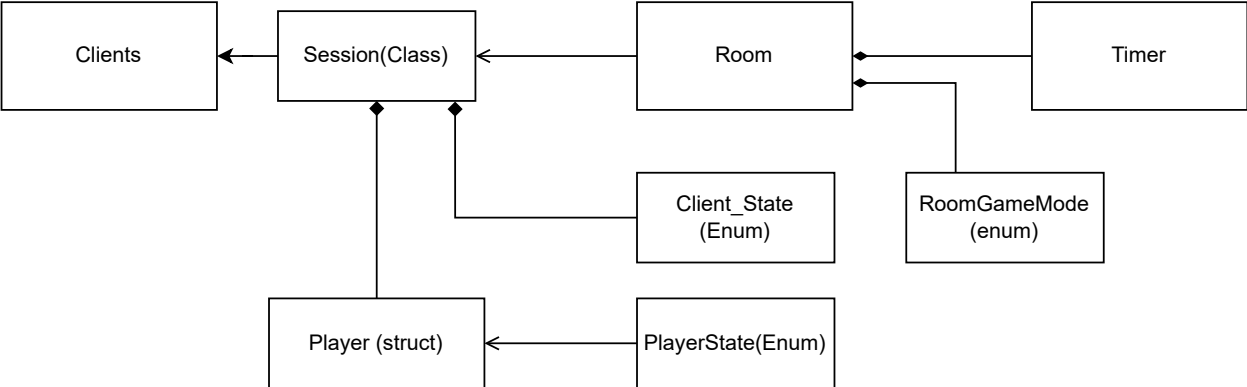


Client High Level Design



Server

Server High Level Design



Client Low Level Design

Stage

```
Stage();
virtual ~Stage();

virtual void Initailizer() = 0;

virtual void Update(float fTime) = 0;

virtual void RestartStage() {};

virtual void LateUpdate(float fTime) = 0;

virtual void Render(HDC mDC) = 0;

virtual void Free() = 0;
```

CStageTutorial : CStage

```
CStageTutorial();
virtual ~CStageTutorial();

virtual void Initailizer() override;

virtual void RestartStage() override;

virtual void Update(float fTime) override;

virtual void LateUpdate(float fTime) override;

virtual void Render(HDC mDC) override;

virtual void Free() override;
```

TimerManager : Singleton

```
TimerManager();
virtual ~TimerManager();

void init();
void Reset();
void Tick();

float GetDeltaTime() const;
float GetFPS() const;

float SetTotalTime(float time);
float GetTotalTime() const;
```

Object

```
Object();
virtual ~Object();

virtual void Initailizer() = 0;

virtual void Update(float fTime) = 0;

virtual void LateUpdate(float fTime) = 0;

virtual void Render(HDC mDC) = 0;

virtual void Free() = 0;

SetPosRect();
GetPosRect();

SetCenterPos();
GetCenterPos();

ResetPosRect();

SetRidius();

GetRidius();

SetDead();

변수
bool m_bDead{};
RECT m_POSRECT{};
POINT m_CenterPos{};
float m_fRidius{};
```

OtherPlayer : Object

```
OtherPlayer();
virtual ~OtherPlayer();

virtual void Initailizer() override;

virtual void Update(float fTime) override;

virtual void LateUpdate(float fTime) override;

virtual void Render(HDC mDC) override;

virtual void Free() override;
```

ObjManager : Object

```
OtherPlayer();
virtual ~OtherPlayer();

virtual void Initailizer() override;

virtual void Update(float fTime) override;

virtual void LateUpdate(float fTime) override;

virtual void Render(HDC mDC) override;

virtual void Free() override;

void SetMyPlayer(CPlayer* _player);

void SetMyPlayerID(int id);

int GetMyPlayerID() const;

COtherPlayer* FindOtherPlayer(int id);

void SetMyPlayer(CPlayer* _player);

void SetMyPlayerID(int id);

int GetMyPlayerID() const;

COtherPlayer* FindOtherPlayer(int id);

void AddOtherPlayer(int id, float x, float y);

void RemoveOtherPlayer(int id);
```

NetworkManager : Singleton

```
NetworkManager();
virtual ~NetworkManager();

bool initialize_Client();

void sendPacket(char* buffer, int size);

void trySendBuffer();

void processPacket();

void updatePacket();

void shutdown();
```

전역 핸들러 함수

```
Handle_AllPlayerMovePacket_s2c

AllPlayer_MovePacket_s2c

Handle_LoginAccept_s2c

Handle_RoomWait_s2c

Handle_RoomStart_s2c

SpawnOtherPlayerPacket_s2c

DespawnOtherPlayerPaccke_s2c
```

<u>MapManager</u>
GetInstance(); DestoryInstance(); MapShow(); MapLoad(); MapStage();

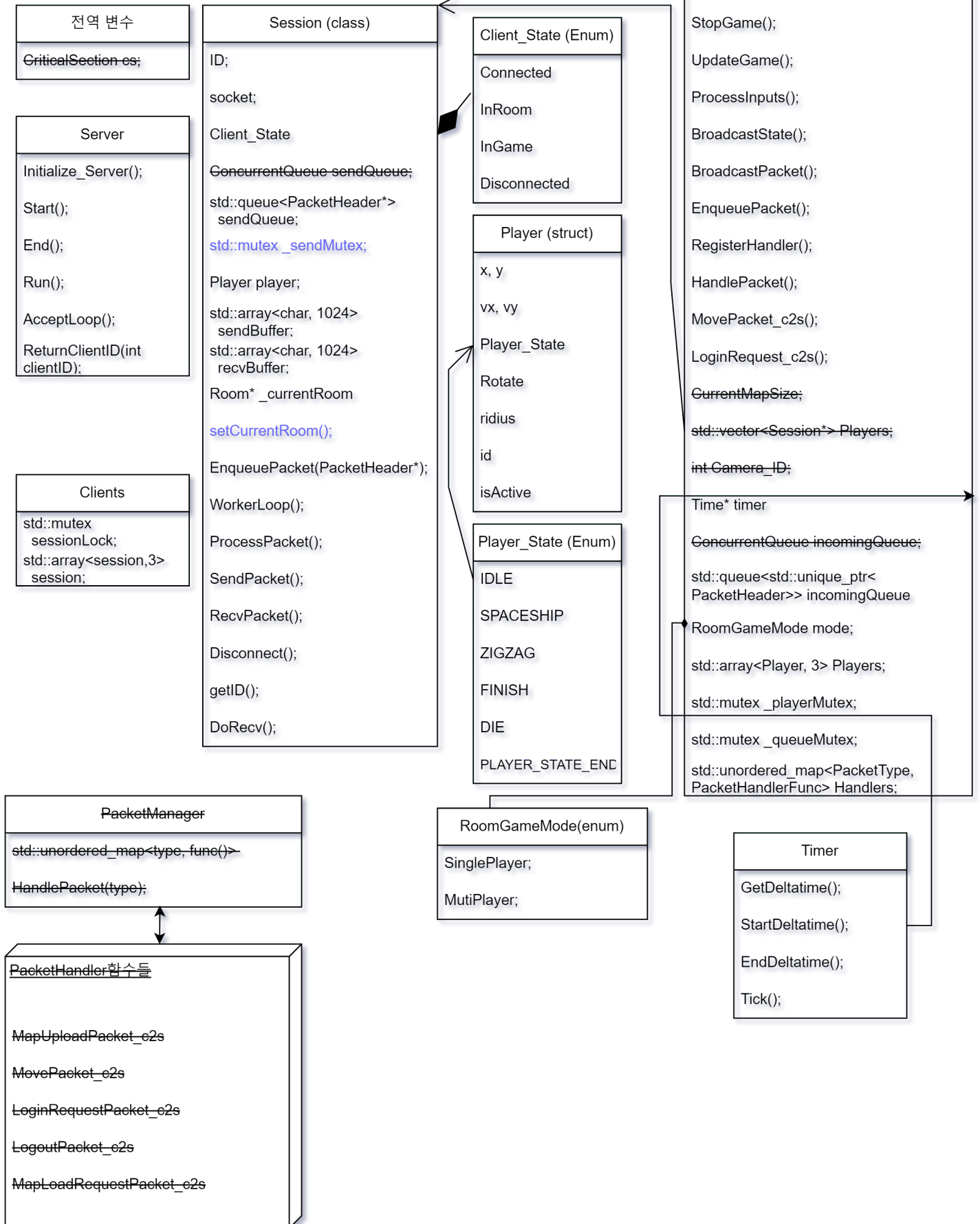
<u>MultiStage : Stage</u>
GetInstance(); DestoryInstance(); MapShow(); MapLoad(); MapStage();

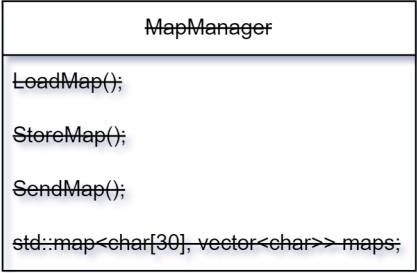
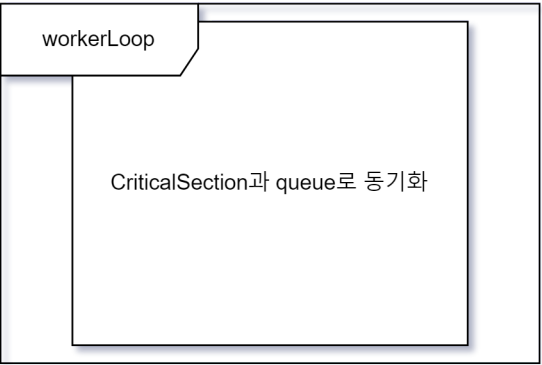
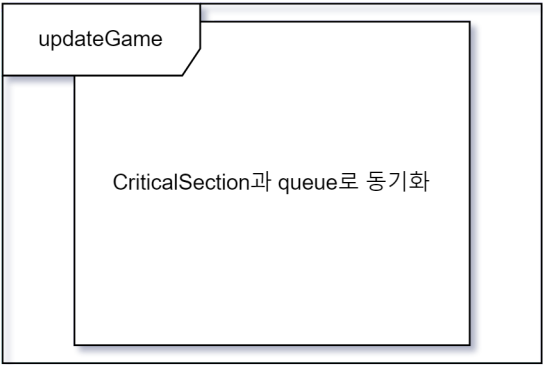
<u>UIManager</u>
GetInstance(); DestoryInstance(); MiniMapUpdate();- MiniMapShow();- ScoreBoardShow();- TimerShow();- UserNumShow();

<u>-PlayerNumUI : Object</u>
PlayerNumUI(); virtual ~PlayerNumUI(); virtual void Initailizer() override; virtual void Update(float fTime) override; virtual void LateUpdate(float fTime) override; virtual void Render(HDC mDC) override; virtual void Free() override;

<u>MiniMapUI : Object</u>
MiniMapUI();- virtual ~MiniMapUI(); virtual void Initailizer() override; virtual void Update(float fTime) override; virtual void LateUpdate(float fTime) override; virtual void Render(HDC mDC) override; virtual void Free() override;

Server Low Level Design





Packet

Client => Server

PacketManager
std::unordered_map<PacketType, func(> Handlers; RegisterHandler(type, func()); Handle(type,session);

Sever => Client

PacketManager
std::unordered_map<PacketType, func(> Handlers; RegisterHandler(type, func()); Handle(type);

MovePacket
long long size; PacketType type int id long x,y; float vx, vy; float rotate float ridius; int cameralD; Player_State player_state; float minMapPercentage;

LogoutPacket
long long size; PacketType type int id;

LoginReqeustPacket
long long size; PacketType type

MovePacket
long long size; PacketType type long x,y; float vx,vy; float rotate Player_State player_state; float minMapPercentage;

s2c_AllPlayerMovePacket
long long size; PacketType type float x[3],y[3]; float vx[3],vy[3]; float ridius[3]; Player_State state[3] float roate[3]

RoomWaitPacket
long long size; PacketType type int playerCount; int playerIDs[MAX_PLAYERS];

RoomStartPacket
long long size; PacketType type int playerCount; int playerIDs[MAX_PLAYERS];

LoginAcceptPacket
long long size; PacketType type int playerID;

SpawnOtherPlayerPacket
long long size; PacketType type int OtherplayerID;

DespawnOtherPlayerPacket
long long size; PacketType type int OtherplayerID;

PacketType(enum class)
MovePacket_s2c AllPlayerMovePacket_s2c MovePacket_c2s LoginAcceptPacket_s2c LoginRequestPacket_c2s LogoutPacket_c2s SpawnOtherPlayerPacket_s2c DespawnOtherPlayerPacket_s2c RoomWaitPacket_s2c RoomStartPacket_s2c

MapLoadRequestPacket
long long size; PacketType type char[30] mapName

MapNamesPacket
long long size; PacketType type std::array<char[30], 10> mapNames

MapLoadPacket
long long size; PacketType type char[30] mapNames vector<char> mapData;

MapUploadPacket
long long size; PacketType type char[30] mapName; vector<char> mapData;

RoomEnterAcceptPacket
long long size; PacketType type

RoomLeavePacket
long long size; PacketType type int roomId;

MapUploadDonePacket
long long size; PacketType type

RoomStartPacket
long long size; PacketType type

RoomEndPacket
long long size; PacketType type int roomId; int winUserID float ElapsedTime